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United States Patent	12406857
Kind Code	B2
Date of Patent	September 02, 2025
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Method for fabricating array structure of columnar capacitor and semiconductor structure

Abstract

Embodiments provide a method for fabricating an array structure of a columnar capacitor and a semiconductor structure. In the method, before a mask layer is removed, a photoresist layer is filled to adjust a thickness of the mask layer in a peripheral region and a thickness of the mask layer in an array region to be equal, thereby preventing a top support layer from being worn due to impacts of different thicknesses of the mask layers on a thickness of the top support layer. In addition, in the method, a third sacrificial layer and an auxiliary layer are further formed to perform dual protection on the top support layer, thereby preventing the top support layer from being thinned in subsequent processes, to increase support strength of the top support layer, thereby further preventing the columnar capacitor from tilting due to insufficient support strength of the top support layer.

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Appl. No.: 18/151469

Filed: January 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230298899 A1	Sep. 21, 2023

Foreign Application Priority Data

CN	202110973338.5	Aug. 24, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/CN2022/072659 20220119 PENDING child-doc US 18151469

Publication Classification

Int. Cl.: H01L21/311 (20060101); H10D1/68 (20250101)

U.S. Cl.:

CPC H01L21/31144 (20130101); H10D1/68 (20250101);

Field of Classification Search

CPC: H10D (1/68); H01G (4/38); H01G (4/33); H01L (21/31144); H10B (12/50)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0077102	12/2016	Kim et al.	N/A	N/A
2018/0166449	12/2017	Pak	N/A	H10B 12/033
2019/0198506	12/2018	Kim	N/A	H10B 12/485
2021/0151439	12/2020	Choi et al.	N/A	N/A
2023/0015120	12/2022	Wan	N/A	H10D 1/68

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103681676	12/2013	CN	N/A
110010604	12/2018	CN	N/A
110504284	12/2018	CN	N/A
111725139	12/2019	CN	N/A
111771279	12/2019	CN	N/A
20120050327	12/2011	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present disclosure is a continuation of PCT/CN2022/072659, filed on Jan. 19, 2022, which claims priority to Chinese Patent Application No. 202110973338.5 titled “METHOD FOR FABRICATING ARRAY STRUCTURE OF COLUMNAR CAPACITOR AND SEMICONDUCTOR STRUCTURE” and filed on Aug. 24, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to the field of semiconductor technology, and more particularly, to a method for fabricating an array structure of a columnar capacitor and a semiconductor structure.

BACKGROUND

(2) As a semiconductor memory device commonly used in computers, dynamic random access memory (DRAM) includes many repeated memory cells. In a process of fabricating the DRAM below 20 nm, the DRAM uses a stacked capacitor structure in most cases, and a capacitor of the DRAM is a vertical columnar capacitor with a high depth-to-width ratio.

(3) Because the columnar capacitor has a high depth-to-width ratio, to increase stability of the columnar capacitor, it is generally necessary to provide a support layer to support the columnar capacitor. A defect in an existing method for fabricating a columnar capacitor is as follows. A top support layer is easily worn. Consequently, support strength of the top support layer is insufficient, and thus the columnar capacitor may be caused to tilt or even peel, which adversely affects performance of the columnar capacitor.

(4) Therefore, it is necessary to provide a method for fabricating an array structure of a columnar capacitor, to resolve a problem in the prior art that a top support layer is easy to be worn, and the like.

SUMMARY

(5) Embodiments of the present disclosure provide a method for fabricating an array structure of a columnar capacitor and a semiconductor structure.

(6) According to some embodiments of the present disclosure, one aspect of the present disclosure provides a method for fabricating an array structure of a columnar capacitor, including:

(7) providing a substrate provided with a plurality of conductive pads therein, where a first sacrificial layer, an intermediate support layer, a second sacrificial layer, a top support layer and a mask layer are stacked on the substrate, the substrate is divided into an array region and a peripheral region, a thickness of the mask layer positioned in the array region is less than a thickness of the mask layer positioned in the peripheral region, and a plurality of capacitor holes in the array region penetrate through the mask layer, the top support layer, the second sacrificial layer, the intermediate support layer and the first sacrificial layer, to expose the plurality of conductive pads; forming a photoresist layer, where the photoresist layer is filled in the plurality of capacitor holes and covers the mask layer in the array region; removing a part of the mask layer in the peripheral region, where an upper surface of a remaining part of the mask layer in the peripheral region is flush with an upper surface of the mask layer in the array region; removing the photoresist layer on a surface of the mask layer and etching the mask layer by using the top support layer as an etching stop layer; forming a third sacrificial layer, where the third sacrificial layer covers the top support layer; removing the photoresist layer; filling a conductive material in the plurality of capacitor holes to form a lower electrode, where the lower electrode is electrically connected to the plurality of conductive pads; forming an auxiliary layer, where the auxiliary layer covers the third sacrificial layer and the lower electrode; patterning the auxiliary layer, the third sacrificial layer and the top support layer, and removing the third sacrificial layer and the second sacrificial layer; patterning the intermediate support layer and removing the first sacrificial layer and the auxiliary layer; forming a dielectric layer, where the dielectric layer covers an exposed surface of the substrate, an exposed surface of the lower electrode, an exposed surface of the intermediate support layer, and an exposed surface of the top support layer; and forming an upper electrode, where the upper electrode covers a surface of the dielectric layer. The step of forming the photoresist layer further includes: forming a photoresist material layer, where the photoresist material layer is filled in the plurality of capacitor holes and covers the mask layer in the array region and the mask layer in the peripheral region; and etching back the photoresist material layer until the mask layer in the

peripheral region is exposed, to form the photoresist layer.

(8) According to some embodiments of the present disclosure, another aspect of the present disclosure further provides a semiconductor structure, which includes: a substrate, where a plurality of conductive pads are arranged in the substrate, and the substrate is divided into an array region and a peripheral region; a first sacrificial layer, an intermediate support layer, a second sacrificial layer, a top support layer and a third sacrificial layer stacked on the substrate, where a surface of the top support layer positioned in the array region is flush with a surface of the top support layer positioned in the peripheral region; a lower electrode arranged in the array region, where the lower electrode penetrates through the third sacrificial layer, the top support layer, the second sacrificial layer, the intermediate support layer and the first sacrificial layer, and is electrically connected to the plurality of conductive pads; and an auxiliary layer, where the auxiliary layer covers the third sacrificial layer and the lower electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To describe the technical solutions of the embodiments of the present disclosure more clearly, the accompanying drawings required for describing the embodiments of the present disclosure will be briefly introduced below. Apparently, the accompanying drawings in the following description are merely some embodiments of the present disclosure. To those of ordinary skills in the art, other accompanying drawings may also be derived from these accompanying drawings without creative efforts.

(2) FIG. 1A is a cross-sectional schematic diagram of forming capacitor holes provided by the first embodiment of the present invention. FIG. 1B is a cross-sectional schematic diagram of a mask layer removed provided by the first embodiment of the present invention. FIG. 1C is a cross-sectional schematic diagram of a conductive material filled provided by the first embodiment of the present invention. FIG. 1D is a cross-sectional schematic diagram of forming a lower electrode provided by the first embodiment of the present invention.

(3) FIG. 2 is a schematic diagram of steps of a method for fabricating an array structure of a columnar capacitor according to a second embodiment of the present disclosure; and

(4) FIG. 3A is a cross-sectional schematic diagram of forming capacitor holes provided by the second embodiment of the present invention;

(5) FIG. 3B is a cross-sectional schematic diagram of filling a photoresist layer in the capacitor holes provided by the second embodiment of the present invention;

(6) FIG. 3C is a cross-sectional schematic diagram of removing partial mask layer of the peripheral area provided by the second embodiment of the present invention;

(7) FIG. 3D is a cross-sectional schematic diagram of exposing the top support layer provided by the second embodiment of the present invention;

(8) FIG. 3E is a cross-sectional schematic diagram of forming a third sacrificial layer provided by the second embodiment of the present invention;

(9) FIG. 3F is a cross-sectional schematic diagram of removing the photoresist layer provided by the second embodiment of the present invention;

(10) FIG. 3G is a cross-sectional schematic diagram of forming a lower electrode provided by the second embodiment of the present invention;

(11) FIG. 3H is a cross-sectional schematic diagram of forming an auxiliary layer provided by the second embodiment of the present invention;

(12) FIG. 3I is a cross-sectional schematic diagram of removing the third sacrificial layer and the second sacrificial layer provided by the second embodiment of the present invention;

(13) FIG. 3J is a cross-sectional schematic diagram of the removing the first sacrificial layer and

the auxiliary layer provided by the second embodiment of the present invention;

(14) FIG. 3K is a cross-sectional schematic diagram of forming medium layer provided by the second embodiment of the present invention;

(15) FIG. 3L is a cross-sectional schematic diagram of forming an upper electrode provided by the second embodiment of the present invention.

DETAILED DESCRIPTION

(16) To make the objectives, technical means and effects of the present disclosure clearer, the present disclosure will be further described below with reference to the accompanying drawings. It should be understood that the embodiments described herein are some but not all of the embodiments of the present disclosure, and are not intended to limit the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present disclosure without creative efforts shall fall within the protection scope of the present disclosure.

(17) FIGS. 1A to 1D are schematic cross-sectional views of a semiconductor structure corresponding to a main process for forming a lower electrode of an array structure of a columnar capacitor according to a first embodiment of the present disclosure.

(18) Referring to FIG. 1A, a substrate **100** is provided. A plurality of conductive pads **101** are arranged in the substrate **100**. A first sacrificial layer **110**, an intermediate support layer **120**, a second sacrificial layer **130**, a top support layer **140** and a mask layer **150** are stacked on the substrate **100**. The substrate **100** is divided into an array region **100A** and a peripheral region **100B**, and a thickness of the mask layer **150** positioned in the array region **100A** is less than that of the mask layer **150** positioned in the peripheral region **100B**. In the array region **100A**, a plurality of capacitor holes **160** penetrate through the mask layer **150**, the top support layer **140**, the second sacrificial layer **130**, the intermediate support layer **120** and the first sacrificial layer **110**, to expose the conductive pads **101**.

(19) Referring to FIG. 1B, the mask layer **150** is removed. In this step, the thickness of the mask layer **150** in the peripheral region **100B** is greater than that of the mask layer **150** in the array region **100A**. Therefore, if the mask layer **150** in the peripheral region **100B** is entirely removed, the top support layer **140** in the array region **100A** is worn, and thus becomes thinner compared with the peripheral region **100B**.

(20) Referring to FIG. 1C, a conductive material **170** is filled. The conductive material **170** is filled in the capacitor hole **160** and covers a surface of the top support layer **140**. In this step, a thickness of the conductive material **170** deposited in the peripheral region **100B** is greater than that of the conductive material **170** deposited in the array region **100A**.

(21) As shown in FIG. 1D, the conductive material **170** on the surface of the top support layer **140** is removed, and a lower electrode **180** is formed in the capacitor hole **160**. In this step, the thickness of the conductive material **170** deposited in the peripheral region **100B** is greater than the thickness of the conductive material deposited in the array region **100A**. Therefore, if the conductive material in the peripheral region **100B** is entirely removed, the thickness of the top support layer **140** in the array region **100A** is further reduced, or even the top support layer **140** in the array region **100A** is entirely removed. Consequently, the subsequently formed columnar capacitor has insufficient top support strength, and thus is prone to tilt, which adversely affects the performance of the array structure of the columnar capacitor, thus further adversely affecting the performance of a memory, and reducing yield of the memory.

(22) To resolve the foregoing technical problem, a second embodiment of the present disclosure further provides a method for fabricating the array structure of the columnar capacitor, which can prevent the top support layer from being worn, thereby ensuring the thickness and the support strength of the top support layer, preventing the columnar capacitor from tilting, improving the performance of the columnar capacitor, and further increasing the yield of the memory. In some embodiments, in the method for fabricating the array structure of the columnar capacitor in the

present disclosure, before the mask layer is removed, the photoresist layer is filled to adjust the thickness of the mask layer in the peripheral region and the thickness of the mask layer in the array region to be equal, thereby preventing the top support layer from being worn due to impacts of different thicknesses of the mask layers on the thickness of the top support layer. In addition, in the method, a third sacrificial layer and an auxiliary layer are further formed to perform dual protection on the top support layer, thereby preventing the top support layer from being thinned in subsequent processes, to increase support strength of the top support layer, thereby further preventing the columnar capacitor from tilting due to insufficient support strength of the top support layer.

(23) FIG. 2 is a schematic diagram of steps of a method for fabricating an array structure of a columnar capacitor according to a second embodiment of the present disclosure. Referring to FIG. 2, the method includes following steps of: Step S20, providing a substrate provided with a plurality of conductive pads therein, where a first sacrificial layer, an intermediate support layer, a second sacrificial layer, a top support layer and a mask layer are stacked on the substrate, the substrate is divided into an array region and a peripheral region, a thickness of the mask layer positioned in the array region is less than a thickness of the mask layer positioned in the peripheral region, and a plurality of capacitor holes in the array region penetrate through the mask layer, the top support layer, the second sacrificial layer, the intermediate support layer and the first sacrificial layer, to expose the plurality of conductive pads; Step S21, forming a photoresist layer, where the photoresist layer is filled in the plurality of capacitor holes and covers the mask layer in the array region; Step S22, removing a part of the mask layer in the peripheral region, where an upper surface of a remaining part of the mask layer in the peripheral region is flush with an upper surface of the mask layer in the array region; Step S23, removing the photoresist layer on a surface of the mask layer and etching the mask layer by using the top support layer as an etching stop layer; Step S24, forming a third sacrificial layer, where the third sacrificial layer covers the top support layer; Step S25, removing the photoresist layer; Step S26, filling a conductive material in the plurality of capacitor holes to form a lower electrode, where the lower electrode is electrically connected to the plurality of conductive pads; Step S27, forming an auxiliary layer, where the auxiliary layer covers the third sacrificial layer and the lower electrode; Step S28, patterning the auxiliary layer, the third sacrificial layer and the top support layer, and removing the third sacrificial layer and the second sacrificial layer; Step S29, patterning the intermediate support layer and removing the first sacrificial layer and the auxiliary layer; Step S30, forming a dielectric layer, where the dielectric layer covers an exposed surface of the substrate, an exposed surface of the lower electrode, an exposed surface of the intermediate support layer, and an exposed surface of the top support layer; and Step S31, forming an upper electrode, where the upper electrode covers a surface of the dielectric layer.

(24) FIGS. 3A to 3L are schematic cross-sectional views of a main semiconductor structure formed by means of the method according to the second embodiment of the present disclosure.

(25) In Step S20, referring to FIG. 3A, a substrate 300 is provided. A plurality of conductive pads 301 are arranged in the substrate 300. A first sacrificial layer 310, an intermediate support layer 320, a second sacrificial layer 330, a top support layer 340 and a mask layer 350 are stacked on the substrate 300. The substrate 300 is divided into an array region 300A and a peripheral region 300B, and a thickness of the mask layer 350 positioned in the array region 300A is less than that of the mask layer 350 positioned in the peripheral region 300B. In the array region 300A, a plurality of capacitor holes 360 penetrate through the mask layer 350, the top support layer 340, the second sacrificial layer 330, the intermediate support layer 320 and the first sacrificial layer 310, to expose the conductive pads 301.

(26) The substrate 300 may include a silicon substrate, a germanium (Ge) substrate, a silicon germanium (SiGe) substrate, a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate, etc. The substrate 300 may also be a substrate including other elemental semiconductors or compound semiconductors, such as gallium arsenide, indium phosphide, or

silicon carbide, etc., and the substrate **300** may also be a stack structure, such as a silicon/germanium silicon stack, etc. In addition, the substrate **300** may be an ion-doped substrate, which may be P-type doped or N-type doped. In the substrate **300**, a plurality of peripheral devices may also be formed, such as field effect transistors, capacitors, inductors and/or P-N junction diodes, etc. In this embodiment, the substrate **300** is the silicon substrate, which further includes other device structures, such as a bit line structure, a transistor structure, etc., which are not shown because they are not related to the present disclosure.

(27) In this embodiment, a material of the first sacrificial layer **310** and a material of the second sacrificial layer **330** may be oxides such as silicon oxide; and a material of the intermediate support layer **320** and a material of the top support layer **340** may be nitrides such as silicon nitride. A material of the mask layer **350** may be polysilicon.

(28) In this embodiment, a bottom support layer **370** is further provided on the substrate **300**. The bottom support layer **370** covers the substrate **300** and exposes the conductive pads **301**, and the first sacrificial layer **310** covers the bottom support layer **370**. A material of the bottom support layer **370** may be nitrides such as silicon nitride.

(29) In Step S21, referring to FIG. 3B, a photoresist layer **380** is formed, where the photoresist layer **380** fills the capacitor hole **360** and covers the mask layer **350** in the array region **300A**. In this embodiment, an upper surface of the photoresist layer **380** is flush with an upper surface of the mask layer **350** in the peripheral region **300B**.

(30) In this step, the photoresist layer **380** is filled into the capacitor hole **360**, and the mask layer **350** in the array region **300A** is protected. In some embodiments, a photoresist material layer is first formed, and then the photoresist material layer is etched back, to form the photoresist layer **380**. In some embodiments, the photoresist material layer is formed, where the photoresist material layer is filled in the capacitor hole **360** and covers the mask layer **350** in the array region **300A** and the mask layer **350** in the peripheral region **300B**. Next, the photoresist material layer is etched back, to expose the mask layer **350** in the peripheral region **300B**, and the remaining photoresist material layer is used as the photoresist layer **380**.

(31) In Step S22, referring to FIG. 3C, in the peripheral region **300B**, part of the mask layer **350** is removed, such that an upper surface of the remaining mask layer **350** in the peripheral region **300B** is flush with that of the mask layer **350** in the array region **300A**.

(32) In this step, the photoresist layer **380** is used as a barrier layer of the mask layer **350** in the array region **300A**, and the mask layer **350** in the peripheral region **300B** is not blocked by the photoresist layer **380**. Therefore, the mask layer **350** in the peripheral region **300B** may be etched and thinned, such that the upper surface of the remaining mask layer **350** in the peripheral region **300B** is flush with the upper surface of the mask layer **350** in the array region **300A**. In some embodiments of the present disclosure, due to limitation of an actual process, when the mask layer **350** in the peripheral region **300B** is removed, the photoresist layer **380** is also partially removed. That is, the photoresist layer **380** is also thinned.

(33) In this step, an etching rate of an etching substance to the mask layer **350** is greater than an etching rate to the photoresist layer **380**, to prevent the photoresist layer **380** from being entirely etched. For example, the mask layer **350** is dry-etched by using at least one of HBr and NF₃ as an etching gas, and the etching rate of the etching gas HBr or NF₃ to the mask layer **350** is greater than the etching rate of the etching gas HBr or NF₃ to the photoresist layer **380**.

(34) An objective of this step is to remove a height difference between the mask layer **350** in the peripheral region **300B** and the mask layer **350** in the array region **300A**. The thickness of the remaining mask layer **350** in the peripheral region **300B** is equal to that of the mask layer **350** in the array region **300A**, such that when the mask layer **350** is subsequently removed, a case (as shown in FIG. 1B) where the top support layer **340** in the array region **300A** is worn to entirely remove the mask layer **350** in the peripheral region **300B** does not occur, thereby preventing the thickness of the top support layer **340** in the array region **300A** from being smaller than that of the

top support layer **340** in the peripheral region **300B**.

(35) In Step S23, referring to FIG. 3D, the photoresist layer **380** on the surface of the mask layer **350** is removed, and the mask layer **350** is etched using the top support layer **340** as an etching stop layer.

(36) In this step, the photoresist layer **380** on the surface of the mask layer **350** is removed to expose the mask layer **350**, and then the mask layer **350** is etched until the top support layer **340** is exposed.

(37) In this step, the mask layer **350** is removed by means of a dry etching process. The etching rate of the etching substance to the mask layer **350** is greater than the etching rate to the photoresist layer **380** and an etching rate to the top support layer **340**, to prevent the photoresist layer **380** and the top support layer **340** from being etched. In this embodiment, the mask layer **350** is a polycrystalline silicon mask layer, and the top support layer **340** is a silicon nitride layer. In this case, the mask layer **350** may be dry-etched by using at least one of HBr and NF.sub.3 as the etching gas, to remove the mask layer **350**. The etching rate of the etching gas HBr or NF.sub.3 to polysilicon is greater than the etching rate of the etching gas HBr or NF.sub.3 to silicon nitride.

(38) In this step, because the thickness of the remaining mask layer **350** in the peripheral region **300B** is equal to the thickness of the mask layer **350** in the array region **300A**, a degree of thinning the top support layer **340** in the peripheral region **300B** is equal to a degree of thinning the top support layer **340** in the array region **300A**, such that after this step, a thickness of an exposed part of the top support layer **340** in the peripheral region **300B** is also equal to a thickness of an exposed part of the top support layer **340** in the array region **300A**.

(39) In Step S24, referring to FIG. 3E, a third sacrificial layer **390** is formed, and the third sacrificial layer **390** covers the top support layer **340**.

(40) In this step, the third sacrificial layer **390** is deposited and formed on the surface of the top support layer **340**. A deposition method may be an atomic layer deposition process, a chemical vapor deposition process, a spin-on deposition process, or the like. In some embodiments, due to limitation of an actual process, the third sacrificial layer **390** not only covers the surface of the top support layer **340**, but also covers the top of the photoresist layer **380**. The third sacrificial layer **390** may be an oxide layer, such as a silicon oxide layer, and the third sacrificial layer **390** may be made of the same material as the first sacrificial layer **310** and the second sacrificial layer **330**.

(41) In some embodiments, a method of forming the third sacrificial layer **390** includes following steps of: forming a third sacrificial material layer, where the third sacrificial material layer covers the top support layer **340** and the photoresist layer **380**; and thinning the third sacrificial material layer until the photoresist layer **380** is exposed, to form the third sacrificial layer **390**. A method of forming the third sacrificial material layer may be an atomic layer deposition process, a chemical vapor deposition process, a spin-on deposition process, or the like.

(42) In Step S25, referring to FIG. 3F, the photoresist layer **380** is removed.

(43) The third sacrificial layer **390** further covers the top of the photoresist layer **380**. Therefore, in this step, the third sacrificial layer **390** is first thinned to expose the photoresist layer **380**, and then the photoresist layer **380** is removed. The photoresist layer **380** may be removed by means of an ashing process.

(44) In Step S26, referring to FIG. 3G, the capacitor hole **360** is filled with a conductive material to form a lower electrode **400**, where the lower electrode **400** is electrically connected to the conductive pad **301**.

(45) The conductive material may be a titanium nitride material or other materials that may be used as the lower electrode of the columnar capacitor. In this step, the conductive material may be deposited by means of an atomic layer deposition process to form the lower electrode **400**. In some embodiments, due to limitation of an actual process, the conductive material is not only filled in the capacitor hole **360**, but also covers a surface of the third sacrificial layer **390** in the array region **300A** and the surface of the third sacrificial layer **390** in the peripheral region **300B**. In this step,

following steps are also included: forming a lower electrode material layer, where the lower electrode material layer not only is filled in the capacitor holes **360**, but also covers the third sacrificial layer **390**; and thinning the lower electrode material layer until the third sacrificial layer **390** is exposed, and rest of the lower electrode material layer serves as the lower electrode **400**. A method of thinning the lower electrode material layer may be etch-back, the third sacrificial layer **390** is exposed, and only the conductive material in the capacitor hole **360** is retained, to form the lower electrode **400**.

(46) In Step S27, referring to FIG. 3H, an auxiliary layer **410** is formed, where the auxiliary layer **410** covers the third sacrificial layer **390** and the lower electrode **400**.

(47) In this step, an auxiliary layer **410** is deposited on the surface of the third sacrificial layer **390** and at a top of the lower electrode **400**. The auxiliary layer **410** may be a nitride layer, for example, a silicon nitride layer. The auxiliary layer **410** and the third sacrificial layer **390** serve as dual protection layers for the top support layer **340**, to prevent the top support layer **340** from being thinned in a subsequent process. In addition, the auxiliary layer **410** may further protect the lower electrode **400** from being damaged in a subsequent process.

(48) In Step S28, referring to FIG. 3I, the auxiliary layer **410**, the third sacrificial layer **390** and the top support layer **340** are patterned, and the third sacrificial layer **390** and the second sacrificial layer **330** are removed.

(49) In some embodiments, in this step, the auxiliary layer **410**, the third sacrificial layer **390** and the top support layer **340** are patterned to form a first opening **341**. Next, the third sacrificial layer **390** and the second sacrificial layer **330** are removed along the first opening **341**, to expose the intermediate support layer **320**. A process for patterning the auxiliary layer **410**, the third sacrificial layer **390** and the top support layer **340** may be photolithography and a dry etching process, and a method of removing the third sacrificial layer **390** and the second sacrificial layer **330** may be a wet etching process.

(50) In Step S29, referring to FIG. 3J, the intermediate support layer **320** is patterned, and the first sacrificial layer **310** and the auxiliary layer **410** are removed.

(51) In some embodiments, in this step, the intermediate support layer **320** is patterned to form a second opening **321**. A position of the second opening **321** corresponds to that of the first opening **341**. A first sacrificial layer **310** is removed along the second opening **321** to expose a substrate **300**. Processes of patterning the intermediate support layer **320** may be a photolithography process and a dry etching process, and a method for removing the first sacrificial layer **310** may be a wet etching process.

(52) In this step, a material of the intermediate support layer **320** is the same as a material of the auxiliary layer **410**. In this case, in the step of patterning the intermediate support layer **320**, the auxiliary layer **410** may be synchronously removed, to expose the top of the lower electrode **400**.

(53) In this embodiment, after the first sacrificial layer **310** is removed, the bottom support layer **370** is exposed.

(54) In Step S30, referring to FIG. 3K, a dielectric layer **420** is formed, where the dielectric layer **420** covers exposed surfaces of the substrate **300**, the lower electrode **400**, the intermediate support layer **320**, and the top support layer **340**.

(55) The dielectric layer **420** may be a high-K dielectric layer to improve the performance of the columnar capacitor. For example, the high-K dielectric layer may be Al_2O_3 , HfO_2 , Ta_2O_5 and ZrO_2 , which may be formed by means of chemical vapor deposition (CVD), atomic layer deposition (ALD), or metal organic chemical vapor deposition (MOCVD), etc.

(56) In Step S31, referring to FIG. 3L, an upper electrode **430** is formed, where the upper electrode **430** covers a surface of the dielectric layer **420**.

(57) In this embodiment, the upper electrode **430** is filled in an interspace among the bottom support layer **370**, the intermediate support layer **320** and the top support layer **340**, and covers the

top support layer **340**. The upper electrode **430**, the dielectric layer **420** and the lower electrode **400** constitute the columnar capacitor. A plurality of columnar capacitors are arranged in an array to constitute an array structure of the columnar capacitor.

(58) In the method for fabricating the array structure of the columnar capacitor in the present disclosure, before the mask layer is removed, the photoresist layer **380** is filled to adjust the thickness of the mask layer **350** in the peripheral region **300B** and the thickness of the mask layer **350** in the array region **300A** to be equal, thereby preventing from causing adverse impacts on the thickness of the top support layer **340** due to different thicknesses of the mask layers **350**. In addition, in the method, the third sacrificial layer **390** and the auxiliary layer **410** are further formed to perform dual protection on the top support layer **340**, thereby preventing the top support layer **340** from being thinned in subsequent processes, to increase the support strength of the top support layer **340**, thereby further preventing the columnar capacitor from tilting due to insufficient support strength of the top support layer **340**.

(59) The present disclosure also provides a semiconductor structure. Referring to FIG. 3H, the semiconductor structure includes a substrate **300**, a plurality of conductive pads **301** are disposed in the substrate **300**, and the substrate **300** is divided into the array region **300A** and the peripheral region **300B**.

(60) The substrate **300** may include a silicon substrate, a germanium (Ge) substrate, a silicon germanium (SiGe) substrate, a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate, etc. The substrate **300** may also be a substrate including other elemental semiconductors or compound semiconductors, such as gallium arsenide, indium phosphide, or silicon carbide, etc., and the substrate **300** may also be a stacked structure, such as a silicon/germanium silicon stack, etc. In addition, the substrate **300** may be an ion-doped substrate, which may be P-type doped or N-type doped. In the substrate **300**, a plurality of peripheral devices may also be formed, such as field effect transistors, capacitors, inductors and/or P-N junction diodes, etc. In this embodiment, the substrate **300** is the silicon substrate, which further includes other device structures, such as a bit line structure, a transistor structure, etc., which are not shown because they are not related to the present disclosure.

(61) The first sacrificial layer **310**, the intermediate support layer **320**, the second sacrificial layer **330**, the top support layer **340** and the third sacrificial layer **390** are stacked on the substrate **300**. The surface of the top support layer **340** positioned in the array region **300A** is flush with the surface of the top support layer **340** positioned in the peripheral region **300B**.

(62) In this embodiment, materials of the first sacrificial layer **310**, the second sacrificial layer **330** and the third sacrificial layer **390** may be oxides, such as silicon oxide. Materials of the intermediate support layer **320** and the top support layer **340** may be nitrides, such as silicon nitride.

(63) In this embodiment, a bottom support layer **370** is further provided on the substrate **300**. The bottom support layer **370** covers the substrate **300** and exposes the conductive pads **301**, and the first sacrificial layer **310** covers the bottom support layer **370**. A material of the bottom support layer **370** may be nitrides such as silicon nitride.

(64) The lower electrode **400** is arranged in the array region **300A**, where the lower electrode **390** penetrates through the third sacrificial layer **390**, the top support layer **340**, the second sacrificial layer **330**, the intermediate support layer **320** and the first sacrificial layer **310**, and the lower electrode **390** is electrically connected to the conductive pad **301**. The lower electrode **400** may be a titanium nitride electrode. A top of the lower electrode **390** may be flush with the surface of the third sacrificial layer **390**.

(65) The auxiliary layer **410** covers the third sacrificial layer **390** and the lower electrode **400**. The auxiliary layer **410** may be a nitride layer, for example, a silicon nitride layer. The auxiliary layer **410** and the third sacrificial layer **390** serve as dual protection layers for the top support layer **340**, to prevent the top support layer **340** from being thinned in a subsequent process. In addition, the

auxiliary layer **410** may further protect the lower electrode **400** from being damaged in a subsequent process.

(66) In the semiconductor structure of the present disclosure, the surface of the top support layer **340** in the array region **300A** is flush with the surface of the top support layer **340** in the peripheral region **300B**. The auxiliary layer **410** and the third sacrificial layer **390** serve as the dual protection layers for the top support layer **340**, to prevent the top support layer **340** from being thinned in the subsequent process. In this way, the thickness and the support strength of the top support layer **340** are ensured, thereby preventing the columnar capacitor formed on the basis of the semiconductor structure from tilting, and thus improving the performance of a memory formed subsequently.

(67) What is mentioned above merely refers to some embodiments of the present disclosure. It shall be pointed out that to those of ordinary skill in the art, various improvements and embellishments may be made without departing from the principle of the present disclosure, and these improvements and embellishments are also deemed to be within the scope of protection of the present disclosure.

Claims

1. A method for fabricating an array structure of a columnar capacitor, comprising: providing a substrate provided with a plurality of conductive pads therein, wherein a first sacrificial layer, an intermediate support layer, a second sacrificial layer, a top support layer and a mask layer are stacked on the substrate, the substrate being divided into an array region and a peripheral region, a thickness of the mask layer positioned in the array region being less than a thickness of the mask layer positioned in the peripheral region, and a plurality of capacitor holes in the array region penetrating through the mask layer, the top support layer, the second sacrificial layer, the intermediate support layer and the first sacrificial layer, to expose the plurality of conductive pads; forming a photoresist layer, wherein the photoresist layer is filled in the plurality of capacitor holes and covers the mask layer in the array region; removing a part of the mask layer in the peripheral region, wherein an upper surface of a remaining part of the mask layer in the peripheral region is flush with an upper surface of the mask layer in the array region; removing the photoresist layer on a surface of the mask layer and etching the mask layer by using the top support layer as an etching stop layer; forming a third sacrificial layer, wherein the third sacrificial layer covers the top support layer; removing the photoresist layer; filling a conductive material in the plurality of capacitor holes to form a lower electrode, wherein the lower electrode is electrically connected to the plurality of conductive pads; forming an auxiliary layer, wherein the auxiliary layer covers the third sacrificial layer and the lower electrode; patterning the auxiliary layer, the third sacrificial layer and the top support layer, and removing the third sacrificial layer and the second sacrificial layer; patterning the intermediate support layer and removing the first sacrificial layer and the auxiliary layer; forming a dielectric layer, wherein the dielectric layer covers an exposed surface of the substrate, an exposed surface of the lower electrode, an exposed surface of the intermediate support layer, and an exposed surface of the top support layer; and forming an upper electrode, wherein the upper electrode covers a surface of the dielectric layer.

2. The method for fabricating the array structure of the columnar capacitor according to claim 1, wherein the forming the photoresist layer further comprises: forming a photoresist material layer, wherein the photoresist material layer is filled in the plurality of capacitor holes and covers the mask layer in the array region and the mask layer in the peripheral region; and etching back the photoresist material layer until the mask layer in the peripheral region is exposed, to form the photoresist layer.

3. The method for fabricating the array structure of the columnar capacitor according to claim 2, wherein further, an upper surface of the photoresist layer is flush with an upper surface of the mask layer in the peripheral region.

4. The method for fabricating the array structure of the columnar capacitor according to claim 1, wherein in the removing the photoresist layer on the surface of the mask layer and etching the mask layer by using the top support layer as the etching stop layer, an etching rate of an etching substance to the mask layer is greater than an etching rate to the top support layer.
 5. The method for fabricating the array structure of the columnar capacitor according to claim 4, wherein further, the mask layer is a polycrystalline silicon layer, and the top support layer is a silicon nitride layer.
 6. The method for fabricating the array structure of the columnar capacitor according to claim 1, wherein in the forming the third sacrificial layer, the third sacrificial layer is formed by means of a spin-on deposition process.
 7. The method for fabricating the array structure of the columnar capacitor according to claim 6, wherein the third sacrificial layer is an oxide layer.
 8. The method for fabricating the array structure of the columnar capacitor according to claim 1, wherein the forming the third sacrificial layer further comprises: forming a third sacrificial material layer, wherein the third sacrificial material layer covers the top support layer and the photoresist layer; and thinning the third sacrificial material layer until the photoresist layer is exposed, to form the third sacrificial layer.
 9. The method for fabricating the array structure of the columnar capacitor according to claim 1, wherein the filling the conductive material in the plurality of capacitor holes to form the lower electrode further comprises: forming a lower electrode material layer, wherein the lower electrode material layer is filled in the plurality of capacitor holes, and covers the third sacrificial layer; and thinning the lower electrode material layer until the third sacrificial layer is exposed, to form the lower electrode.
 10. The method for fabricating the array structure of the columnar capacitor according to claim 1, wherein the patterning the auxiliary layer, the third sacrificial layer and the top support layer, and removing the third sacrificial layer and the second sacrificial layer further comprises: patterning the auxiliary layer, the third sacrificial layer and the top support layer to form a first opening, wherein the first opening exposes the third sacrificial layer and the second sacrificial layer; and removing the third sacrificial layer and the second sacrificial layer along the first opening to expose the intermediate support layer.
 11. The method for fabricating the array structure of the columnar capacitor according to claim 10, wherein the patterning the intermediate support layer and removing the first sacrificial layer and the auxiliary layer further comprises: patterning the intermediate support layer to form a second opening; removing the first sacrificial layer along the second opening to expose the substrate; and removing the auxiliary layer.
 12. The method for fabricating the array structure of the columnar capacitor according to claim 11, wherein a position of the first opening corresponds to a position of the second opening.
 13. The method for fabricating the array structure of the columnar capacitor according to claim 1, further comprising a bottom support layer, wherein the bottom support layer covers the substrate, the conductive pad is exposed, and after the patterning the intermediate support layer and removing the first sacrificial layer, the bottom support layer is exposed.
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